SECS v2.3

School-Based Exam Control System

Comprehensive Operational User Manual & Guide for Markers, KP Moderators and HODs

Aligned with Singapore Ministry of Education (MOE) Primary School Paper 1 Standards | 100% Offline & Client-Side

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1. System Overview & Quick Start

The **School-Based Exam Control System (SECS)** is an offline, client-side web application designed specifically for primary school Paper 1 (Writing) marking. It completely eliminates error-prone shared spreadsheets, version conflicts, accidental formula overrides, and clerical calculation errors while safeguarding student data privacy under the Public Sector (Data Protection) framework.

1.1 Launching the System

1. Navigate to the system folder: `school-exam-control-system`.
2. Double-click **`start_secs.bat`** (or right-click **`index.html`** → Open with Microsoft Edge or Google Chrome).
3. The application runs completely offline in your browser. **No internet connection, web server, or software installation is required.**

1.2 Role Access & Security Matrix

To preserve examination segregation of duties, the system enforces distinct security tiers:

Portal Role	Credential Type	Primary Capabilities	Data Privacy Safeguard
Exam Marker	Paper Marker Passcode (Customized per paper)	Select target Exam Paper, rapid score entry with Score Confirmation Modal, Green/Pink qualitative feedback tags, personal and class marking progress bars.	Strictly shielded: Cannot view peer markers' scores, moderation deltas, or cohort-wide statistics.
KP Moderator (Senior Teacher / LH)	Paper Marker Passcode + **KP Authorization PIN / OTP** (2-Step Verification)	Standard marking capabilities + direct access to the ⭐ **KP Moderation & Script Sampling Hub** within the Marker Portal for standardisation sampling and score moderation after secondary PIN verification.	Shielded from altering rubric presets or deleting cohorts; moderation audit trail recorded under their designated marker profile.
HOD / Exam Lead	HOD Master Passcode (Confidential to HOD)	Multi-Exam Registry (P3–P6), Roster import (CSV with Chinese characters), rubric configuration, live cohort consolidation, sample check audit & moderation, 1-click marking freeze, auto-save exit guard, item analysis, passcode/PIN management, and differentiated exports.	Full cohort oversight, permanent initial score retention, variance logging, and marker calibration analytics.

2. Marker Operational Guide (For Exam Markers)

The Marker Portal is streamlined for rapid, distraction-free score entry while grading physical hardcopy composition scripts.

```
[1] Login (Paper Marker Passcode) ──► [2] Select Target Exam Paper & Marker Code
[2B] KP 2-Step PIN / OTP (If KP Moderator Selected) ──► [3] Select Class Bundle
[4] Check Real-Time Class Progress Card ──► [5] Enter Content & Language Scores
[6] Tag Authentic Feedback (🟢 Green Strengths & 🔴 Pink Gaps)
[7] Click "💾 Save & Next" ──► [8] Verify Score Confirmation Modal (Enter ↵)
```

2.1 Real-Time Marked Script Tracking (Cohort & Class Level)

Markers have immediate, real-time visibility over their grading progress across three levels:

- **Top Header Badges**:
  - 👥 **Cohort Marked**: Shows total cohort completion progress, e.g. `150/200 marked (75.0%)`.
  - 🏫 **Class Marked**: Shows the selected class bundle's progress, e.g. `24/28 marked (85.7%)`.
  - 📝 **My Marked**: Shows your personal cumulative marked count across all assigned classes.
- **Dropdown Live Annotations**: When selecting a class in the dropdown, every class option displays its live progress tally (e.g., `Humility 1 (Primary 4.1) [24/28 marked]`).
- **Live Class Progress Card (Left Panel)**:
  - Animated progress bar showing the exact percentage of scripts completed for the active class.
  - **Physical Stack Remaining Counter**: Displays exactly how many hardcopy papers remain in your physical stack (e.g., `4 papers remaining in physical bundle`), eliminating manual paper recounting.

2.2 Locating Physical Scripts

1. Choose your filtering view: **Form Class** (e.g. `Humility 1 (Primary 4.1)`) or **Mother Tongue Class** (e.g. `4MT-1`).
2. Select the class bundle matching the physical stack in front of you.
3. Use the **Instant Bilingual Search Bar** to locate any candidate immediately by typing:
   - Register Number (e.g., `5` or `#05`)
   - English Name (e.g., `Wei Ting` or `Tan`)
   - Chinese Characters (e.g., 陈 or 韦廷)
   - Script ID (e.g., `SC-005`)
4. Click on the student card in the roster to open their score entry drawer.

2.3 Scoring & Boundary Guards

- **Content Mark**: Enter marks awarded for plot development, idea elaboration, and theme relevance.
- **Language Mark**: Enter marks awarded for vocabulary, sentence variation, paragraphing, and mechanics.
- **Automatic Sum & Percentage**: The system calculates the total and percentage instantly upon keystroke.
- **Real-Time Input Guard**: If a mark exceeds the prescribed rubric maximum (e.g., entering 9 when Content max is 7), the field flashes red with an alert blocking erroneous entry.

2.4 Authentic Qualitative Feedback Tagging (Content vs Language)

Feedback descriptors are strictly aligned with MOE Primary Paper 1 (Writing) rubrics, divided into **Content** and **Language** criteria, and color-coded for intuitive selection:

- 🟢 **Green Descriptors (写作亮点与优势 / Writing Strengths)**: Highlight commendable techniques and authentic literary merit.
- 🔴 **Pink Descriptors (常见失分点与学习短板 / Common Mistakes & Gaps)**: Identify diagnostic learning gaps for subsequent classroom remediation.

Criterion	🟢 Writing Strengths (Highlights)	🔴 Common Mistakes & Weaknesses
1. Content Evaluation 内容评价与反馈	紧扣图意或题目 · 结构清楚 · 内容生动 · 故事精彩 · 详略写分明 · 事件结果合理 · 结尾扣题 · 立意积极深刻 · 描写细腻生动 · 过渡自然	离题 · 偏题 · 内容不完整 · 内容平淡 · 事件交代不清楚 · 事件结果不合理 · 事件结果不完整 · 结尾没点题 · 详略不当 · 缺乏心理与细节描写 · 结构凌乱 · 分段不清 · 篇幅不足
2. Language Evaluation 语言与表达评价与反馈	词汇丰富 · 句式有变化 · 好词好句丰富 · 标点准确·字迹工整	人称错误 · 句式重复 · 句式语病多 · 用词不精准 · 搭配不当 · 受方言或英文思维影响 · 标点符号错误 · 错别字较多

2.5 High-Speed Marking Flow (Save & Next)

- Click the prominent blue 💾 **`Save & Next Script`** button.
- Scores and tags save instantly to local storage in under 5 milliseconds.
- The system automatically advances to the next unmarked script in your current class bundle, enabling continuous two-handed grading.

2.6 Pre-Advance Score Confirmation Dialog (Keyboard Optimized)

To eliminate clerical errors when grading high volumes of physical paper scripts, the system prompts markers with an instant verification dialog before advancing:

- **Comprehensive Score Summary**: Displays Candidate Name, Class, Script ID, Content score, Language score, Total, Percentage, and all selected qualitative tags.
- **Keyboard Shortcut: Enter to Confirm**: The "Confirm & Next" button is automatically focused. Simply press **`Enter`** ↵ on your keyboard to save and immediately load the next script.
- **Keyboard Shortcut: Esc to Amend**: Press **`Escape (Esc)`** or click "Amend / Cancel" to return focus directly to score editing.

2.7 Understanding Marking Lock (Read-Only Mode)

> 🔒 **Read-Only Marking Freeze**: When the HOD locks the marking session after the deadline:
> - An amber banner displays: "🔒 *Marking is currently FROZEN by HOD. The system is in READ-ONLY mode for review.*"
> - Score inputs, feedback tags, and the Save button are disabled to prevent inadvertent alterations.
> - Markers can still view and review all their graded scripts, marks, and remarks.

2.8 Key Personnel (KP) Moderation & Script Sampling Hub

Teachers designated as **Key Personnel (KP)** (e.g. Senior Teachers, Subject Heads, Level Heads) have direct access to standardisation and moderation tools directly from their Marker Portal:

- **Dual-Tab Navigation**: Authorized KP markers see a navigation bar at the top with two dedicated tabs:
  - 🖍️ `My Marking Entry`: Standard physical script scoring interface for their assigned bundle.
  - ⭐ `KP Moderation & Script Sampling Hub`: Full paper-wide moderation workspace, equipped with a live badge showing total recommended scripts awaiting check.
- **View All Scripts Across All Markers**: KP moderators can inspect every candidate script registered under the active paper, including unmarked scripts, marked scripts, and previously moderated scripts across all teachers.
- **Intelligent Standardisation Sampling Recommendations**: One-click filter pills instantly identify scripts meeting official quality audit triggers:
  - 🎯 **Recommended Combined** : Prioritised composite queue encompassing all critical quality triggers.
  - ⚖️ **Borderline Pass (48% - 52%)** : Scripts hovering immediately above or below the 50% passing threshold.
  - 🌟 **Borderline Quality (78% - 82%)** : Scripts nearing distinction/quality cut-off.
  - ⚡ **Content vs Language Discrepancy** : Unbalanced scripts with >20% variance between Content and Language criteria.
  - 🎲 **Random 10% Sample** : Representative cross-section for general standardization audit.
  - 🚨 **Pupils-at-Risk (PAR < 50%)** : Rapid identification of candidates requiring learning support.
  - ⏳ **Unmarked Scripts** : Immediate visibility into unentered scripts to monitor cohort pace.
- **Side-by-Side Moderation Drawer**:
  - **Original Marker Evidence**: Displays First Marker Code, initial Content score, Language score, Total, Percentage, awarded Feedback Tags (Green/Pink), and Teacher Notes.
  - **Moderation Controls**: KP enters Moderated Content and Language scores. The system calculates real-time **Variance Delta (±Δ)** with color cues (green for agreement, amber/red for adjustments).
  - **Moderation Rationale**: Mandatory justification log ensures transparent audit trails aligned with standardisation meeting guidelines.

3. HOD & Key Personnel Operational Guide

As HOD or Examination Lead (authenticated via confidential **HOD Master Passcode**), you have complete administrative control over multiple examination levels, active papers, archived papers, cohort class bundles, rubrics, sample check moderation, and official reports.

3.1 HOD Authentication & Executive Paper Control Deck

- **Executive Paper Control Deck**: Pinned permanently across the top of the HOD interface:
  - **Active Paper Metadata**: Level, Subject, Component, Academic Year, and Rubric Max bounds (e.g., `15 Max (7C + 8L)`).
  - **Status Badges**: Real-time indicators for 🟢 **Active Marking**, 🔒 **Marking Frozen**, and 📦 **Archived Paper**.
  - **Live Candidature & Marked Tracking**: Displays total candidate count and live percentage completion (e.g., `150 / 200 (75%)`).
  - **Quick Actions**: 1-click access to 📁 **Papers & Archives**, 🖍️ **Edit Paper Settings**, 🔒 **Lock/Unlock Marking**, and 📦 **Archive Paper**.

3.2 Unified Exam Studio & Cohort Manager (Tab 1)

SECS unifies paper configuration, class bundle management, and candidate rosters into a single master-detail workspace:

- **Left Rail — Paper Navigator & Archived Papers**:
  - **Active Papers List**: Interactive cards displaying candidate tallies, marked progress bars, and marker team size. Click "*Switch To*" to instantly switch paper context. Supports 1-click archiving, duplicating, and editing.
  - **Archived Papers**: Safely stores concluded exam cycles in read-only freeze. Retains complete candidate scores and audit logs with 1-click restoration to active marking anytime.
  - **Register New Paper**: Inline creation form with Level, Subject, Component, Rubric Presets, and 1-click "*Copy roster from existing paper*".
- **Right Rail — Active Paper Setup & Cohort Workspace**:
  - **Step 1: Paper Specifications**: Configure Examination Title, Level, Subject, and Rubric Presets (e.g., P3/P4 MT 7C+8L=15 vs P5/P6 MT 20C+20L=40).
  - **Step 2: Candidate Roster Management**: Candidate table with class-level dropdown filtering (`rosterClassFilter`), real-time search (by Register No, English/Chinese name, Script ID), 1-click cross-paper roster cloning, and bilingual CSV roster upload with UTF-8 BOM protection.
  - **Step 3: Marker Team & KP Moderation**: Allocate markers (M01-M20) and check `KP Moderator` to delegate sample check and moderation rights without sharing the master HOD password.
  - **Step 4: Paper Security & Passcodes**: Set paper-specific marker passcode (isolating cohorts) and configure the confidential KP Moderator Authorization PIN / OTP (e.g. `MOD88`) with an optional 1-click random PIN generator. Both passcodes are prominently summarized on each paper card in the Papers Hub.
  - **Class & Cohort Bundles (Bottom Section)**: Visual card grid for all Form Classes and Mother Tongue Banded Classes in this paper. Supports 🖍️ `Rename Class`, 🗑️ `Delete Class Bundle`, `+ Add New Class Bundle`, and individual `+ Pupil` additions.

📋 **How to Add Rosters & Class Bundles to Different Exam Papers (4 Seamless Workflows):**

1. **Method 1: 1-Click "Copy from Paper" (Cross-Paper Roster / Class Clone) — [RECOMMENDED]**: In the Right Rail Step 2, click 📋 `Copy from Paper`. Select any source paper, choose **"All Classes"** or a specific **Form Class / MT Class**, select Append or Replace mode, and click 📋 `Copy Pupils Now`. Candidates are imported with clean, unmarked status and sequential script numbers without re-uploading spreadsheets.
2. **Method 2: Inherit Roster upon Registering a New Paper**: In the Left Rail (Register New Paper tab), select `Student Roster & Classes: Copy roster from [Existing Paper]`. The new paper is immediately created with the full cohort and class structure ready for marking.
3. **Method 3: Switch Active Paper & Add Locally**: Switch to the desired paper via the Left Rail or top Exam Selector, and use `+ Add New Class Bundle`, `+ Add Student`, or 📥 `Upload Roster` directly in the Right Rail.
4. **Method 4: Duplicate an Entire Exam Paper**: In the Left Rail or Manage Registry, click 📋 `Duplicate` to clone the entire paper (cohort, rubric, and markers) with scores reset to unmarked.

3.3 Live Monitoring: Master Consolidation (Tab 2)

- **Dynamic Tracking Banner**: Top banner shows real-time cohort statistics: **`Total Marked: X/Y (Z%)`**, **`Unmarked: N`**, and **`Moderated: M`**.
- **Contextual Subgroup Progress**: When filtering by a specific class or marker, the banner dynamically highlights that subgroup's completion (e.g., `Selected Class: 4 Harmony [24/28 marked, 4 remaining]`).
- **Filter Dropdown Counts**: All filter dropdowns annotate each selection with real-time marked counts.
- **Dual-Mark & Moderator Code Display**: Moderated scripts prominently display the final mark alongside an audit badge: `14 (Orig: 13, +1)`, and both original marker and moderator code (e.g., `Mod: M01` or `Mod: HOD`) are clearly shown for total traceability.

3.4 Quality Assurance: HOD Sample Check & Moderation (Tab 3)

Standardised Examination Moderation Protocol:

1. **Smart Sampling Filters**: Automatically flags target scripts: Borderline Pass (48% - 52%), Borderline Quality Pass (78% - 82%), Content/Language Discrepancy, and Random 10% Sample.
2. **Side-by-Side Moderation Drawer**: Inspect the first marker's scores, tags, and qualitative comments, enter moderated scores, and observe the live variance delta ($\pm\Delta$). Both First Marker code and Moderator Code are attributed in the table.
3. **Indelible Historical Retention**: Applying moderation updates the final score while **permanently retaining the first marker's original score** and moderation rationale for audit defense.

3.5 Independent Per-Paper Marking Lock / Read-Only Freeze Control

SECS provides **granular, per-paper marking control**, allowing the HOD to freeze or open individual assessment papers independently:

- **Independent Paper Scheduling**: The HOD can freeze Primary 4 when grading concludes, while keeping Primary 5 actively open for marking. Different assessment levels never interfere with each other.
- **4 Visual Paper Badges**: 🟢 **Active Marking** , 🔒 **Marking Frozen** , ⚠️ **Draft (Unreleased)** , and 📦 **Archived** .
- **1-Click Freeze Toggles**: Toggle freeze directly on each paper card in the Left Rail (🔒 `Freeze` / 🔓 `Open`) or via the header button (🔒 `Freeze Marking`) above the active workspace.

- **Instant Shared Folder Broadcast**: Lock state updates broadcast to `exam_master_${examId}.json` within seconds, placing markers into read-only review mode automatically.

3.6 Item Analysis & Marker Calibration (Tabs 4 & 5)

- **Tab 4 (Item Analysis)**: Cohort Mean, Median, Standard Deviation (σ), Min/Max spread, Quality Pass Rate ($\ge 80\%$), Pass Rate ($\ge 50\%$), and Pupils-at-Risk (PAR $< 50\%$).
- **Tab 5 (Marker Calibration)**: Evaluates inter-rater reliability across all markers. Identifies grading tendencies: Calibrated (± 1.5 marks), Lenient/Dove, Strict/Hawk, and tracks HOD sample checks and moderation deltas.

3.7 Differentiated & Configurable Reports (Tab 6: Export Hub)

SECS generates tailored reports designed for specific stakeholders:

Feature	Option A: Class Teacher Diagnostic Report	Option B: Comprehensive HOD Audit Report
Target Audience	Form Teachers, Subject Teachers, Level Teachers	HOD, Year Head, Vice-Principal, Principal
Scope	Single Class (Form or MT Banded)	Full Cohort or Filtered Class
Marker Reliability Math	**Omitted** (Keeps report clean & pedagogically focused)	**Included** (Marker calibration, Hawk/Dove index, moderation audit)
Pupil Roster	Includes Register No, English Name, Chinese Name, Initial Mark, Final Mark, Delta (Δ), Tags, and Remarks	Full cohort roster with complete moderation audit trails
Pedagogical Interventions	Action plan for Pupils-at-Risk ($< 50\%$) and Quality Pass Bridging ($65\% - 79\%$)	Department-wide strategic recommendations & grade distribution
Configurability	Standardized 1-click generation	**8 Interactive Checkboxes** to customize exact report sections
File Formats	Excel (`.xlsx`) and Microsoft Word (`.doc`)	Excel (`.xlsx`) and Microsoft Word (`.doc`)

3.8 Key Personnel (KP) Moderation Rights

- **Decentralized Standardization**: HODs can delegate sample checking and moderation privileges to senior teachers or Level Heads without sharing the elevated master HOD passcode.
- **1-Click Assignment**: In **Tab 1: Exam Studio & Cohort Manager** (Step 3: Marker Allocation), check the `KP Moderator` box next to any marker.
- **Dedicated Moderation Hub**: When the marker logs in with their standard passcode, they receive a ⭐ **KP Moderator** badge and immediate access to the ⭐ **KP Moderation & Script Sampling Hub**.

3.9 Live Auto-Save Beacon & Exit Verification Guard

- **Continuous Auto-Save Pulse**: A live pulsing badge in the top navigation bar displays the exact time of the latest auto-save (e.g. `Auto-saved 14:02:15`).
- **Exit Verification Guard**: Clicking the Lock / Exit button triggers an exit confirmation modal summarizing all saved cohorts, candidature counts, and marked scripts, with a 1-click option to download a complete JSON database backup file.

3.10 Database Backup, Transfer & Restore

- All exam data is maintained in the browser's local storage sandbox.
- **Backup / Export**: Click 💾 **`Backup DB (JSON)`** to save a snapshot named `SECS_Exam_Database_Backup_YYYY-MM-DD.json`.
- **Restore / Transfer to Another Laptop**:
  1. Copy the backup JSON file and the `school-exam-control-system` folder to the new laptop.
  2. Open SECS in Edge or Chrome.

3. Click 📂 **`Restore DB (JSON)`** and choose your backup file. All students, scores, tags, and settings are restored in seconds.

3.11 Multi-Paper & Multi-Marker Shared Folder Live-Sync Guide (Platinum Standard)

> **📁 Concurrent Multi-Device Marking via School Shared Drive (e.g. S:\...):**
>
> SECS features a **Decoupled Shared Folder Live-Sync Engine** using Microsoft Edge's native File System Access API. Each teacher auto-saves to their own file (`secs_sync/marker_${examId}_${markerCode}.json`), and moderators save to partitioned files (`secs_sync/moderation_${modBy}.json`), preventing file-locking collisions even when 20+ teachers and multiple moderators mark simultaneously across multiple papers.
>
> - **HOD Setup & Automatic Sync**: Place the SECS folder on the school shared drive (`S:\Departments\...`). Click 📁 `Link Shared Folder` in the top navigation bar. Any change made to student rosters, class bundles, or rubrics automatically synchronizes to the exam master file in the background.
> - **Marker Workflow**: Markers open `index.html` from the shared folder, select their target assessment paper (e.g. P4 or P5), enter their Marker Code and PIN, and grade. Marks auto-save silently to their individual JSON file.
> - **KP Moderator Live Queue & Partitioned Moderation**: Multiple Key Personnel and the HOD moderate concurrently without conflicts, auto-saving to separate `moderation_${modBy}.json` files in real time.
> - **Zero Shared Folder Maintenance**: Archiving papers automatically hides them from active queues while preserving complete historical files on disk. **No manual file deletion is ever needed.**

4. Frequently Asked Questions (FAQ)

Q1: Can we use SECS for more than 5 form classes?

Yes, absolutely. SECS dynamically adapts to any number of classes (e.g. 6, 8, 10, 15, or more classes). The roster, dropdowns, and comparison tables generate automatically based on the class names present in your CSV file.

Q2: Can I move the system to a USB thumb drive or my Desktop?

Yes. You must copy or move the **entire `school-exam-control-system` folder**, because `index.html` references scripts in the `js/` folder. Alternatively, you can create a shortcut to `start_secs.bat` on your Desktop without moving any files.

Q3: Why do Chinese characters display properly without corrupted symbols (mojibake)?

SECS includes explicit **UTF-8 Byte Order Marks (BOM,)** and Microsoft Word HTML headers across all CSV, Excel, and Word exports. This guarantees that Chinese characters (如: 陈韦廷, 蔡恩熙) render flawlessly in Microsoft Excel and Word on Windows.

Q4: Can markers see marks or feedback entered by other teachers?

No. The Marker Portal is strictly sandboxed. Markers only see their own assigned scripts, input controls, and personal grading history. Peer marks, cohort statistics, and moderation deltas require the confidential HOD master passcode or authorized KP moderation rights.

Q5: What happens if an error is found after marking has been frozen?

The HOD simply clicks 🔓 **`Unlock Marking`** in the top header or paper card. The marker can then amend the score, save the change, and the HOD can re-freeze the system.

Q6: Can different markers mark concurrently from different laptops in a shared folder?

Yes! Using the **Shared Folder Live-Sync Engine**, multiple markers open `index.html` from the school shared network drive (`S:\`) on their respective laptops. Each marker auto-saves to their own file (`marker_${examId}_${markerCode}.json`), eliminating file collisions, while HOD and KP moderators view live incoming marks and score tallies in real time.

Q7: Do you recommend storing different papers (e.g. P4 and P5) in the same shared folder or different folders?

We strongly recommend storing them in the same shared folder (e.g. `S:\Departments\MotherTongue\Exam2026\`). SECS uses paper-isolated filenames (`exam_master_${examId}.json`, `marker_${examId}_${markerCode}.json`) and a unified manifest. Storing them in a single folder allows markers and moderators to switch between papers seamlessly without re-linking shared drives, while ensuring 100% data separation.

Q8: Can two papers be active at the same time?

Yes. Multiple papers can be active simultaneously (e.g. P4 Paper 1 and P5 Paper 1). Markers grading P4 and markers grading P5 can grade concurrently on their own laptops. HOD can also freeze P4 once finished, while P5 continues marking uninterrupted.

Q9: If an exam paper is archived, do we need to delete any files from the shared folder?

Strictly NO. Archiving automatically updates the manifest to hide the paper from active marking queues, while retaining all historical candidate records and moderation logs on disk for audit compliance. No manual file management is ever required on the shared folder.